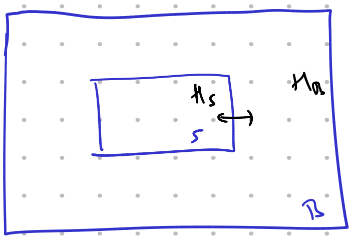


$$H = \overbrace{H_S + H_B}^{H_0} + \lambda H_I$$

$$\rho_S(t) = \mathrm{Tr}_B(\rho(t))$$

$$U_0(t, t_0) = e^{i H_0 (t - t_0)}$$

U ...

$$\rho(t) = U_0^\dagger U \rho(0) U^\dagger U_0$$

$$H_I(t) = U_0^\dagger H_I U_0$$

EOM.: $\frac{d}{dt}\rho = -i\left[H_I(t), \rho(t)\right]$

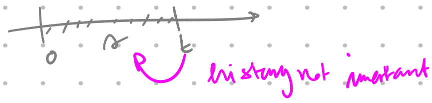
$$\rho(t) = \rho(0) - i\int_0^t dt' \left[H_I(t'), \rho(t')\right]$$

$$\rho_S(t) = \mathrm{Tr}_B(\rho(t))$$

$$\frac{d}{dt}\rho_S = \mathrm{Tr}_B\left((-i)^2 \int_0^t ds \left[H_I(t), \left[H_I(s), \rho(s)\right]\right]\right)$$

assume:

1. sys << bath : $\rho_S(s) \otimes \rho_B$
2. $\tau_B \sim$ correlation
   $\tau_B << \tau_S$ (Born-Markov approx).
3. $\rho_S(t) \otimes \rho_B$: Redfield? ← zamenjaj $s \to t$ (lokalno v času)



4. integracijske meje: $\int_0^t \to \int_0^\infty ds$ $\quad$ $-\infty \quad t-s$

→ Markov Approx: $\frac{d}{dt}\rho_S(t) = -\int_0^\infty ds\, \mathrm{Tr}_B\left(\left[H_I(t), \left[H_I(t-s), \rho_S(t) \otimes \rho_B\right]\right]\right)$

5. Secular approx: (ni oscilacij)

→ $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$ → $\sum_\omega A_\alpha(\omega) = \sum_\omega \sum_{\varepsilon - \varepsilon' = \omega} \pi(\varepsilon) A_\alpha \pi(\varepsilon')$ $\quad$ ω ε' ε H_S A_α

A_α: syst. $\quad$ B_α: Bath

↓ time evol.:

$$H_I(t) = \sum_{\alpha\omega} e^{-i\omega t} A_\alpha(\omega) \otimes e^{iH_B t} B_\alpha e^{iH_0 t}$$

$$\dot{\rho}_S = \sum_{\omega, \omega'} \sum_{\alpha\beta} e^{i(\omega - \omega')t}\, \Gamma_{\alpha\beta}(\omega, t) \left(A_\alpha(\omega)\rho_S(t) A^\dagger_\alpha(\omega') - A^\dagger_\alpha(\omega') A_\beta(\omega)\rho_S(t)\right) + h.c.$$

Secular ($\omega = \omega'$)

↓ single freq. driving, ni pa za jenene z termalno porazdelitvijo!

$\mathrm{Tr}_B(\ldots)$

ne vem kje je Tr_B ... $\Big\}$: $\Gamma_{\alpha\beta}(\omega, t) = \int_0^\infty ds\, \mathrm{Tr}_B\left(B^\dagger_\alpha(t) B_\beta(t-s)\rho_B\right) =$

$$= \frac{1}{2}\underbrace{\gamma_{\alpha\beta}}_{>0} + i S_{\alpha\beta}(\omega)$$

$$\to \dot{\rho}_S = -i\left[H_{LS}, \rho_S(t)\right] + \hat{D}(\rho_S(t))$$

↓

$$H_{LS} = \sum_\omega \sum_{\alpha\beta} S_{\alpha\beta}(\omega)\, A_\alpha^\dagger(\omega) A_\beta(\omega)$$

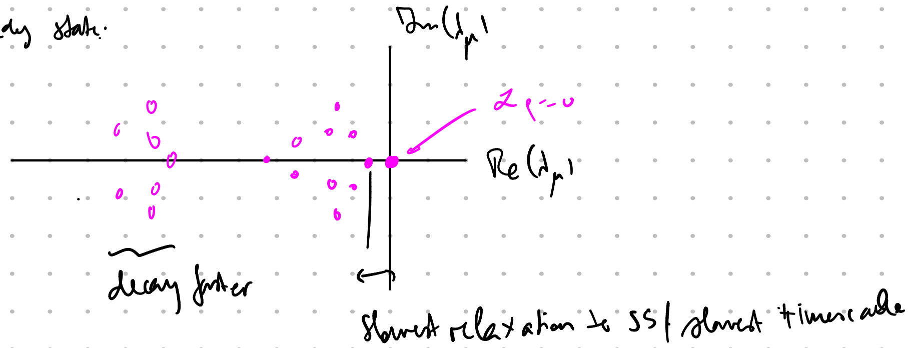
$$\hat{\mathcal{D}} = \sum_\omega \sum_{\alpha\beta} \underbrace{\gamma_{\alpha\beta}(\omega)}_{>0} \left(A_\beta(\omega) \rho_s A_\alpha(\omega) - \tfrac{1}{2}\{A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_s\} \right) \overset{!}{=}$$ diagonaliziramo v $\alpha\beta$, ω?

$$\boxed{= \underbrace{\sum_k \gamma_k \left(L_k \rho_s L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho_s\} \right)}_{\hat{\mathcal{D}}} \qquad \gamma_k > 0 .}$$

→ Schrodinger picture:

$$\boxed{\frac{d}{dt}\rho_s(t) = -i\left[H_s + H_{LS}, \rho_s(t)\right] + \hat{\mathcal{D}}(\rho_s(t)) = \hat{\mathcal{L}}\rho_s(t)}$$

Spekter $\hat{\mathcal{L}}$: $\quad \hat{\mathcal{L}}\rho = 0$ steady state.



Steady state:

$$\underset{v}{\mathbb{1}\,\mathcal{L}} = 0 = \mathcal{L}^\dagger(\mathbb{1}) = \sum A_k^\dagger A_k - \tfrac{1}{2}\{A_k^\dagger A_k, A_k A\} = 0 \quad \checkmark$$

→ $\exists\ \rho_{NESS}$: $\quad \hat{\mathcal{L}}\rho_{NESS} = 0 \qquad$ NESS: Nonez. leq. steady state.

$$\langle \mathbb{1} | \rho_{NESS} \rangle = \mathrm{Tr}(\rho_{NESS}) = 1 .$$

$$\langle \mathbb{1} | u^R_{\mu \neq NESS} \rangle = 0 \ \leftarrow$$ decaying modes, not NESS

Numerika:

ED $\quad \mathcal{L} u^R_\mu = \lambda_\mu u^R_\mu$

→ vektorizacija ρ: $\ n\times n \to n^2 \times 1 \quad$:→ Ex. $\rho = \begin{bmatrix} a_{11} & a_{10} \\ a_{01} & a_{00} \end{bmatrix} \to |\rho\rangle = \begin{bmatrix} a_{11} \\ a_{10} \\ a_{01} \\ a_{00} \end{bmatrix}$

→ matrix multip.: $\quad \mathrm{vec}(ABC) = A \otimes C^T \mathrm{vec}(B)$ ⊛

→ scalar prod.: $\quad \mathrm{tr}(A^\dagger B) = \mathrm{vec}(A)^\dagger \mathrm{vec}(B) = (\ \cdots\ ^*) \begin{pmatrix} \vdots \\ \vdots \end{pmatrix}$

→ **EOM:** $\frac{d}{dt}\rho = \mathcal{L}\rho = -i[H,\rho] + \sum_k L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\}$ (*)

$$\frac{d}{dt}|\rho\rangle = \Big(-i \underset{\text{ket}}{H\otimes 1} + i\, \underset{\text{bra}}{1\otimes H^T} + \sum_k L_k\otimes (L_k^\dagger)^T - \frac{1}{2}\big(L_k^\dagger L_k\otimes 1 + 1\otimes (L_k^\dagger L_k)^T\big)\Big)|\rho\rangle$$

$$\boxed{\frac{d}{dt}|\rho\rangle = \Big(-i\, H\otimes 1 + i\, 1\otimes H^T + \sum_k L_k\otimes L_k^* - \frac{1}{2}(L_k^\dagger L_k\otimes 1) - \frac{1}{2}(1\otimes L_k^T L_k^*)\Big)|\rho\rangle}$$

(*) $ABC = A\otimes C^T \mathrm{vec}B \leftrightarrow L_k\rho L_k^\dagger = (L_k\otimes (L_k^\dagger)^T)|\rho\rangle$

$\{L_k^\dagger L_k, \rho\} \to \overset{A}{1}\,\overset{B}{\rho}\,\overset{C}{L_k^\dagger L_k} = (1\otimes (L_k^\dagger L_k)^T)|\rho\rangle$

Expect. values: $\mathrm{Tr}(\hat{O}\rho) = \langle 1|\hat{O}\otimes 1|\rho\rangle$

$\mathrm{Tr}(\rho) = \langle 1|\rho\rangle = 1.$

Perturbative approach:

$\mathcal{L}\rho = -i[H,\rho] + \varepsilon\hat{\mathcal{D}}\rho \quad \to \quad \rho = \rho_0 + \varepsilon\rho_1 + \varepsilon^2\rho_2 + \dots$

0th. order: $\mathcal{L}\rho_0 = -i[H,\rho_0] = 0$ steady state

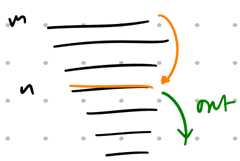
↳ $[\rho_0, H] = 0 \leftrightarrow \underbrace{\rho_0 = \sum_n p_n|n\rangle\langle n| \qquad H|n\rangle = E_n|n\rangle}_{\text{ni novih informacij o } p_n.}$

↳ šele $\hat{\mathcal{D}}$ določa koeficiente p_n.

$\langle n|\dot{\rho}_0|n\rangle = \dot{p}_n = \langle n|\varepsilon\hat{\mathcal{D}}\rho_0|n\rangle =$ $\quad 1 = \sum_k |k\rangle\langle k|$

$= \varepsilon\langle n|\sum_\mu L_\mu \sum_m p_m|m\rangle\langle m|L_\mu^\dagger - \frac{1}{2}\{L_\mu^\dagger L_\mu, \sum_m p_m|m\rangle\langle m|\}|n\rangle =$

$$= \varepsilon\underbrace{\sum_m\sum_\mu |\langle n|L_\mu|m\rangle|^2 p_m}_{= R_{nm},\ m\to n} - \varepsilon\underbrace{\sum_k\sum_\mu |\langle k|L_\mu|n\rangle|^2 p_n}_{\text{out: } = R_{kn},\ n\to k}$$



$k\to m$ rename

$$\dot{p}_n = \varepsilon\sum_m\big(\underbrace{R_{nm}p_m}_{\text{transition v } n.} - R_{mn}p_n\big)$$

R_{mn}: transition iz: $n\to m$ (ven iz n)

$\vec{p}_{NESS}$: kernel matrike: $\underline{D}$, $\underline{D}\,\vec{p}_{NESS} = 0$

$$\underline{D}\,\vec{p}_{NESS} = \begin{bmatrix} -\sum_m R_{1m} & R_{12} & R_{13} & \cdots \\ R_{21} & -\sum_m R_{2m} & R_{23} & \cdots \\ \vdots & \vdots & \ddots & \end{bmatrix}\begin{bmatrix} p_1\\ p_2\\ p_3\\ \vdots\end{bmatrix}$$

$2^N \times 2^N$ ✓

diagonala: [illegible] $n=1$. etc.

$$\dot{p}_1 = -\sum_m R_{m1} p_1 + \sum_m R_{1m} p_m$$

$$\begin{bmatrix} -\sum R_{m1} & R_{12} & R_{13} & \cdots \\ \cdots & & & \end{bmatrix}\begin{bmatrix} p_1 \\ \vdots \end{bmatrix} =$$

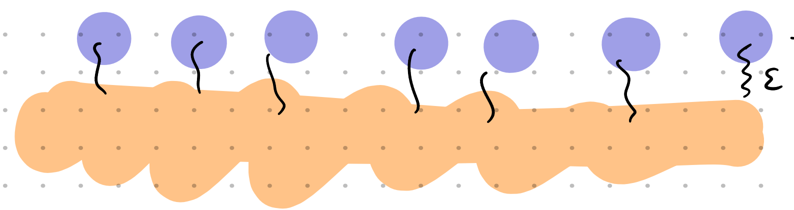
$$= -\sum R_{m1} p_1 + R_{12}p_2 + R_{13}p_3 + R_{14}p_4 \cdots$$

higher orders:

$$0 = \mathcal{L}\rho = (\mathcal{L}_0 + \varepsilon\mathcal{D})(\rho_0 + \delta\rho)$$

$$\delta\rho = -\mathcal{L}^{-1}\varepsilon\hat{\mathcal{D}}\rho_0 = \lim_{\eta\to 0}\left(\mathcal{L}_0 - \eta\,\mathbb{1} + \varepsilon\mathcal{D}\right)^{-1}\varepsilon\,\mathcal{D}\rho_0$$

$\vdots$



$\leftarrow$ E conserved

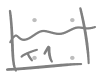
$\}\ \varepsilon$

$$\rho_0 = \sum_m p_m |m\rangle\langle m| \longrightarrow \frac{e^{-\beta H}}{Z} \quad \Big|\ \beta = ?$$

kdaj gre to a thermal state, pri kateri temperaturi?

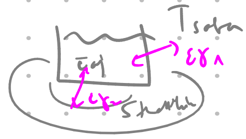
a) thermal bath coupling

npr. fermioni.



T_2 : $\rightarrow$ ekvilibracija v T_2

b) non-thermal bath:



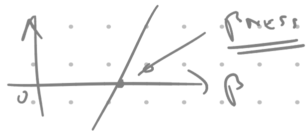
$\Big\}\ \beta \sim f\left(\frac{\gamma_2}{\gamma_1}\right)$

↳ a) $\frac{R_{mn}}{R_{nm}} = e^{-\beta(E_m - E_n)}$; $\beta \sim \frac{1}{T_2}$ (detailed balance)

↳ $\vec{p}_{NESS} \Rightarrow p_n = \frac{e^{-\beta E_n}}{Z}$ $\Big\}$ logično ✓

↳ b) $\frac{R_{mn}}{R_{nm}} \neq e^{-\beta(E_m - E_n)} \Rightarrow$ find β from $\langle \dot{H} \rangle \overset{!}{=} 0$!

$$\langle \dot{H} \rangle = \varepsilon\,\mathrm{Tr}\left(H\left(\gamma_1\mathcal{D}_1\left(\frac{e^{-\beta H}}{Z}\right) + \gamma_2\mathcal{D}_2\left(\frac{e^{-\beta H}}{Z}\right)\right)\right) \overset{!}{=} 0$$



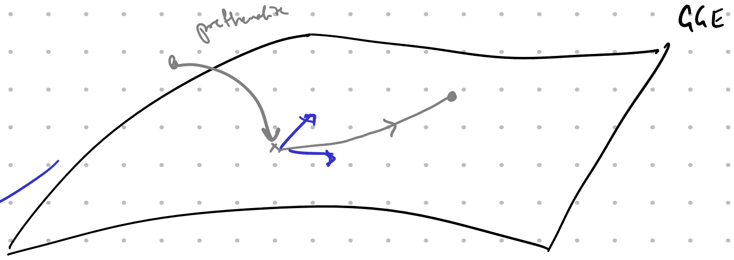
$[H, C_i] = 0$

$C_i =$	same H	$\{C_i\} > 1$
	$\rho_{NESS} = \frac{e^{-\beta H}}{Z} + \delta\rho$	$\rho_{NESS} = \frac{e^{-\sum_i \lambda_i C_i}}{Z} + \delta\rho(\varepsilon)$
	$\beta : \mathrm{Tr}\left(H \varepsilon \mathcal{D}\left(\frac{e^{-\beta H}}{Z}\right)\right) = 0$	$\lambda_i : \underline{\mathrm{Tr}\left(C_i \mathcal{D} e^{-\sum_j \lambda_j C_j}\right)} \overset{!}{=} 0$

Dynamics for $|\psi_0\rangle$ $(i c)$; $\rho(0) = |\psi_0\rangle\langle\psi_0|$

↳ prethermalize to a GGE : $\rho_{GGE}(0)$; $\lambda_i(0) = \langle\psi_0|C_i|\psi_0\rangle = \mathrm{Tr}\left(C_i \frac{e^{-\sum_j \lambda_j C_j}}{Z}\right)$

↳ later times:



definiramo: $\hat{P}_\lambda X = \left.\frac{\partial \rho_{GGE}}{\partial \lambda_i}\right|_{\pi} \chi_{ij}^{-1} \mathrm{Tr}(C_j X)$; $\chi_{ij} = -\mathrm{Tr}\left(C_i \frac{\partial \rho_{GGE}}{\partial \lambda_j}\right)$

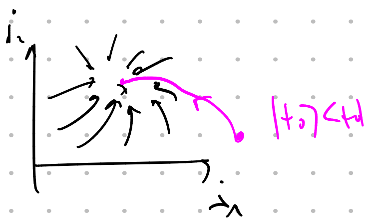
projektor.

time dependence: $\lambda_j(t)$

$$P_\lambda(\dot{\rho}_0) = \sum_i \frac{\partial \rho_{GGE}}{\partial \lambda_i} \dot{\lambda}_i = \sum_j \frac{\partial \rho_{GGE}}{\partial \lambda_i} \chi_{ij}^{-1} \mathrm{Tr}\left(C_i \varepsilon \hat{\mathcal{D}} \rho_{GGE}\right) \Rightarrow$$

$\mathcal{D}$ evolution $\sim$ NESS.

$\rightarrow$



GGEs | Iris:

→ nonintegrable systems: GE: $\langle A(t)\rangle \overset{t\to\infty}{=} \mathrm{Tr}(A\rho_\beta)$; $\rho_\beta = \frac{1}{Z}e^{-\beta H}$,

lahko tudi $\rho = \rho_{\beta,\mu} = e^{-\beta H - \mu N}$, če se število delcev ohranja

initial state (ψ_0) določi končno β in μ:

→ $\langle\psi_0|H|\psi_0\rangle \overset{!}{=} \mathrm{Tr}(H\rho_\beta)$.

→ integrabilni sistemi: $[H, C_i] = 0$ za makroskopsko količino C_i

↳ $\langle\psi_0|C_i|\psi_0\rangle \neq \mathrm{Tr}(C_i\rho_{GE})$!

$$\langle A(t\to\infty)\rangle = \mathrm{Tr}(A\rho_{GGE}) \qquad \boxed{\rho_{GGE} = \rho_{\vec\lambda} = \frac{1}{Z}e^{-\sum_i \lambda_i C_i}}$$

drugi za podsisteme

→ λ_i fiksiramo z: $\underline{\langle\psi_0|C_i|\psi_0\rangle \overset{!}{=} \mathrm{Tr}(C_i\rho_{GGE})}$

→ Weakly odprimo sistem:

or GGE of H_0. Weak dissipation, however, permits a separation of timescales analogous to prethermalization: the dominant unitary dynamics rapidly establishes local equilibrium, while the environment changes the conserved quantities much more slowly.

→ transient prethermalization under H_0 (noninteracting):

↳ pripelje na GGE / GE manifold

↳ disipacija → počasna evolucija $\vec\lambda(t)$ do fiksne točke, ki jo lahko določi od GGE!

→ $\frac{d\rho}{dt} = \hat{\mathcal{L}}\rho = -i[H_0,\rho] + \varepsilon\mathcal{D}\rho$, $[H_0, C_i] = 0$; $\rho = \rho_0 + \delta\rho$

disipator jih zlomi

tudi applicable to GE / $\rho_{\beta,\mu}$.

→ lokalno relaksiran sistem: $\rho_0 = \rho_{\lambda_\infty} \propto \exp\left[-\sum_i \lambda_i(\infty)\, C_i\right]$

↳ stacio stanje

$$0 = \langle\dot C_i\rangle = \mathrm{Tr}[C_i\dot\rho] = \mathrm{Tr}[C_i\hat{\mathcal{L}}\rho] \simeq \varepsilon\,\mathrm{Tr}[C_i\hat{\mathcal{D}}\rho_0]$$

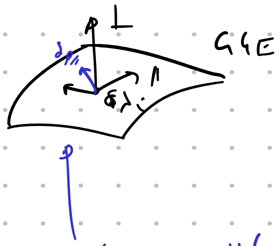
Pogoj:

$$\boxed{\langle\dot C_i\rangle_{\lambda(\infty)} = \mathrm{Tr}[C_i\hat{\mathcal{D}}\rho_0] = 0 \quad \forall \lambda_i.}$$

$\mathcal{E}$ ne vpliva na SS, vpliva na $\tau \sim \frac{1}{\varepsilon}$.

↳ posledično: weak pumping → $\mathcal{O}(1)$ response.

→ popravki $\delta\rho$ nad SS: $\delta_\rho = -\varepsilon \hat{\mathcal{L}}^{-1}\hat{\mathcal{D}}\rho_0$; $\mathcal{L}^{-1} = \lim_{\eta\to 0}(\mathcal{L} - \eta \mathbb{1})^{-1}$

→ 

→ lažko pišemo in hitro smeri:

$\delta\rho_{\parallel} = \sum_i \frac{\partial \rho_\lambda}{\partial \lambda_i} \delta\lambda_i$

Projektor na $\parallel$ (tangentni prostor): $P X = -\sum_{ij} \frac{\partial \rho_\lambda}{\partial \lambda_i} (\chi^{-1})_{ij} \mathrm{Tr}[C_j X]$

$\chi_{ij} = \langle C_i C_j \rangle_{\vec\lambda} - \langle C_i \rangle_{\vec\lambda} \langle C_j \rangle_{\vec\lambda} = -\mathrm{Tr}\left[C_i \frac{\partial \rho_{\vec\lambda}}{\partial \lambda_j}\right]$

?

$\langle \cdot \rangle_{\vec\lambda} = \mathrm{Tr}[\cdot \, \rho_{\vec\lambda}]$

→ $\delta\rho_\parallel$ leži v GGE in popravi λ_i

→ $\delta\rho_\perp = (1 - \hat{P})\delta\rho$ leži izven → hitra dinamika!

→ $\rho(t) = \rho_{\lambda(t)} + \delta\rho(t)$ → na daljši skali blizu GGE manifolda

$\dot\rho = \mathcal{L}\rho \quad / P$

$P\dot\rho \simeq \sum_i \frac{\partial \rho_{\lambda_i}}{\partial \lambda_i} \dot\lambda_i$

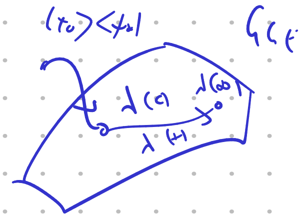
$\parallel$

$-\sum_{ij} \frac{\partial \rho_\lambda}{\partial \lambda_i} (\chi^{-1})_{ij} \underbrace{\mathrm{Tr}[C_j \dot\rho]}_{\varepsilon \mathrm{Tr}[C_j \mathcal{D}\rho_\lambda]}$ →

EOM za λ_i

$$\underbrace{\dot\lambda_i}_{\varepsilon F_i(\vec\lambda)} = -\varepsilon \sum_j (\chi^{-1})_{ij} \mathrm{Tr}[C_j \mathcal{D}\rho_{\vec\lambda(t)}]$$

slow time $\tau = \varepsilon t$: $\boxed{\frac{d\lambda}{d\tau} = F(\lambda)}$, fixed point: $F(\lambda_\infty) = 0$



→ i.c.: $\langle \psi_0 | C_i | \psi_0 \rangle = \mathrm{Tr}[C_i \rho_{\lambda(0)}] \to \lambda(0)$

↓

$\lambda(t) \ldots$

— Rezultat: $\rho_{\vec{\lambda}} \to \rho_{\vec{\mu}}$; $\lambda_i c_i \to \mu_k n_k$

$\langle \dot{c}_i \rangle = \ldots \to$ $\langle \dot{n}_k \rangle(t) = \mathrm{Tr}[n_k \mathcal{D} \rho_\mu] = \varepsilon \sum_j \langle L_j^\dagger n_k L_j \rangle_{\vec{\mu}(t)} - \langle L_j^\dagger L_j n_k \rangle_{\vec{\mu}(t)}$

$\langle \cdot \rangle_\mu = \mathrm{Tr}[\cdot \, \rho_\mu]$

moramo transformirati $L_j \to$ k bazo, Wick,

⋮ naredi sam! (L_j i-t dem-h!)

$\langle \dot{n}_k \rangle \sim \sum_{k'} \langle 1 - n_k \rangle \langle n_{k'} \rangle + \ldots$ itd. itd.! (scattering eq. za kvazidelce)

in $\chi_{ij} \to$ diagonalen:

$\chi_{qq} = \frac{e^{\mu_q(t)}}{(1+e^{\mu_q(t)})^2}$; $\dot{\mu}_q(t) = -\chi_{qq}^{-1}(t) \langle \dot{n}_q \rangle(t)$